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EXPERIMENT NO. - 10

Aim :- Mini Project: FinPredict - Decentralized Financial Planning & Analysis (FP&A) with Time-Series Forecasting and Linear Programming Optimization

Project Details

Problem statement

Personal and small-business financial planning frequently suffers from spending volatility, irregular discretionary surges, and forward cash-flow insolvency. Conventional spreadsheet templates and bookkeeping tools are strictly reactive—they record where capital was spent in the past, but fail to forecast future cash-flow deficits or solve for mathematically optimal spending adjustments.

Furthermore, commercial cloud-based budgeting services require users to upload unencrypted bank statements and Personally Identifiable Information (PII) to remote multi-tenant servers, creating substantial privacy risks under data protection regulations (such as the Indian Digital Personal Data Protection Act, 2023).

The engineering objective of this mini-project is to build **FinPredict**, a decentralized decision-support platform that:

1. Ingests user income and categorized historical expenses in Indian Rupees (₹).
2. Predicts future cash-flow trajectories up to 90 days ahead using time-series forecasting.
3. Automatically computes mathematically optimal budget rebalancing under non-negotiable living survival floors using Linear Programming.
4. Detects statistical spending anomalies in real-time.
5. Enforces a 100% client-side Zero-Trust air-gapped architecture with cryptographic PII masking.

Dataset Identification

The system models structured personal and departmental budget envelopes standardized in Indian Rupees (₹). The data schema consists of nine core numerical attributes capturing inflows, essential survival floors, and elastic discretionary envelopes:

Attribute / Feature	Feature Role & Meaning	Data Type	Constraint & Typical Range (₹)	Living Floor Status
`SALARY_INCOME`	Net monthly post-tax earnings or budget inflow	Numerical	$\geq$ ₹5,000 (Default: ₹85,000)	Primary Inflow

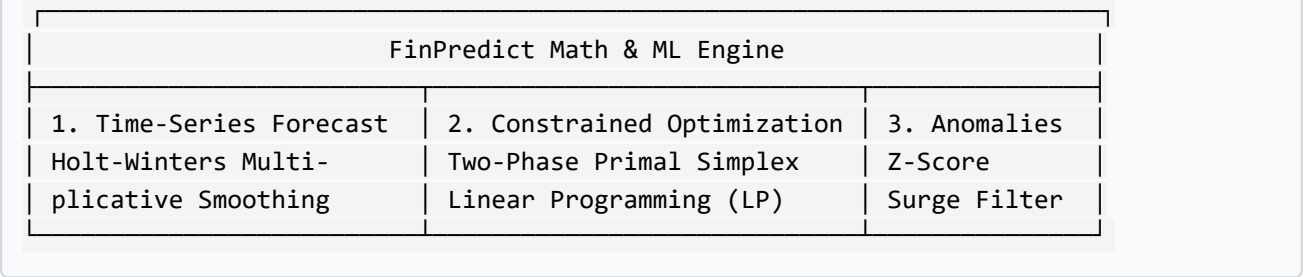
Attribute / Feature	Feature Role & Meaning	Data Type	Constraint & Typical Range (₹)	Living Floor Status
`HOUSING`	Rent, mortgage EMI, society maintenance	Numerical	≥ 20 of Salary	Hard Essential Floor
`FOOD`	Groceries, staples, essential daily nutrition	Numerical	≥ 12 of Salary	Hard Essential Floor
`TRANSPORT`	Fuel, public transit passes, vehicle upkeep	Numerical	≥ 4 of Salary	Semi-Elastic Outflow
`UTILITIES`	Electricity, water, cooking gas, broadband	Numerical	≥ 4 of Salary	Hard Essential Floor
`HEALTHCARE`	Doctor visits, prescription medicines, insurance	Numerical	≥ 4 of Salary	Hard Essential Floor
`ENTERTAINMENT`	OTT subscriptions, dining out, recreation	Numerical	Floor ≥ 0	Fully Elastic Discretionary
`SHOPPING`	Apparel, consumer electronics, retail items	Numerical	Floor ≥ 0	Fully Elastic Discretionary
`OTHER`	Miscellaneous unplanned discretionary outflow	Numerical	Floor ≥ 0	Fully Elastic Discretionary

Data Quality & Validation Rules:

- **Sanitization:** Rejects non-numeric tokens, null values, and negative expenditure entries.
- **Indian Numbering Normalization:** Standardizes inputs to `en-IN` numbering conventions (Lakhs and Crores).
- **Solvency Constraint Check:** Validates that total essential floors do not exceed 100% of declared net income before running mathematical routines.
- **Missing Value Treatment:** Missing transaction intervals are forward-filled using the median value of the preceding 7-day cyclical window.

ML Algorithms

FinPredict integrates three complementary mathematical and machine learning algorithms:



1. Holt-Winters Multiplicative Triple Exponential Smoothing

- **Purpose:** Forecasts future daily and monthly expenditure trajectories across a 30 to 90-day time horizon.
- **Mathematical Rationale:** Financial outflow dynamics exhibit three concurrent components:
  i. **Level (ℓ_t):** Baseline volume of transactions.
  ii. **Trend (b_t):** Long-term upward drift (inflation, lifestyle creep).
  iii. **Seasonality (s_t):** Monthly cyclical spikes occurring on paydays (1st of month for rent/EMI) and weekend surges, governed by periodicity $L = 30$ days.
- **Governing Formulations:**

$$\ell_t = \alpha \frac{y_t}{s_{t-L}} + (1-\alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}$$

$$s_t = \gamma \frac{y_t}{\ell_t} + (1-\gamma)s_{t-L}$$

$$\hat{y}_{t+h} = (\ell_t + h \cdot b_t) \cdot s_{t-L+((h-1) \bmod L) + 1}$$

(Smoothing parameters: $\alpha = 0.2$, $\beta = 0.1$, $\gamma = 0.3$, $L = 30$).

- **Confidence Intervals:** 95% upper and lower prediction bounds are computed using Root Mean Square Error:

$$\text{Bound}_{t+h} = \hat{y}_{t+h} \pm 1.96 \cdot \text{RMSE} \cdot \sqrt{h}$$

2. Two-Phase Primal Simplex Linear Programming (LP)

- **Purpose:** Determines the exact rupee reduction needed in each discretionary envelope to satisfy a user-defined monthly savings target while strictly safeguarding essential survival expenses.
- **Linear Program Formulation:**

$$\min \quad Z = \sum_{j \in D} c_j \cdot (x_j^{\text{current}} - x_j^{\text{optimal}})$$

$$\text{Subject to:} \quad \sum_{j=1}^{n} x_j \leq \text{Income} - \text{Target Savings}$$

$$x_j \geq \text{Floor}_j \quad \forall j \in \text{Housing, Food, Utilities, Healthcare}$$

$$x_j \geq 0 \quad \forall j$$

- **Two-Phase Algorithm:**
  - **Phase 1:** Introduces artificial slack variables to prove feasibility. If the feasible region is empty (e.g., target savings forces total spend below minimum living survival floors), Phase 1 terminates and flags insolvency.
  - **Phase 2:** Executes Gauss-Jordan tableau pivoting to find the optimal rebalancing vector that minimizes discretionary lifestyle reduction.

3. Z-Score Statistical Anomaly Detector

- **Purpose:** Identifies sudden expenditure spikes and unusual transactions.
- **Formulation:**

$$Z = \frac{x_i - \mu_{30}}{\sigma_{30}}$$

- Transactions with $|Z| > 1.96$ or exceeding 35% variance against the 30-day moving median are automatically classified as financial anomalies.

Why Not Deep Learning (LSTM / Transformers)?

1. **Mathematical Explainability:** Auditing requires deterministic proofs. Simplex LP provides exact constraint tableaus; neural networks are non-transparent black boxes.
2. **Zero-Latency In-Browser Execution:** Holt-Winters and Simplex converge in < 5 ms in browser memory without requiring GPU infrastructure.
3. **Zero Data Leakage:** In-browser mathematical execution guarantees financial ledgers never leave the client device.

Proposed System

The proposed system adopts a decoupled, five-layer decentralized architecture:

1. **Presentation & Observation Layer:** React 19 single-page application with responsive Tailwind CSS. Includes the interactive Observation Controller, role-based dashboards (User Analyst vs. Admin Compliance), and SVG cash-flow charts.
2. **Air-Gapped Mathematical Layer:** Client-side execution of Holt-Winters time-series smoothing, Two-Phase Simplex LP solver, and Z-score anomaly detector. Operates with zero outbound network egress.
3. **Security & PII Masking Shield:** Real-time tokenization of sensitive identifiers (Aadhaar, PAN, bank accounts) into secure tokens (`***-XX-****`).
4. **Cloud Database Layer:** Node.js Express REST API connected to MongoDB Atlas storing encrypted user credentials, budget category schemas, and tamper-evident audit logs.
5. **Certified Reporting Layer:** Automated generation of GAAP/Ind-AS P&L statements, Balance Sheets, and certified PDF/Excel exports.

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Figure 1: System Architecture and Decentralized 5-Layer Dataflow Pipeline

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Figure 2: User Dashboard with Interactive Salary & Expense Observation Controller

Implementation

Technology Stack:

- **Frontend:** React 19, TypeScript, Vite, Tailwind CSS, Lucide Icons, Recharts.
- **Backend:** Node.js, Express.js REST API.
- **Database:** MongoDB Atlas Cloud with Mongoose ODM.
- **Math / Analytics:** Pure TypeScript in-memory implementations of Holt-Winters and Primal Simplex LP.
- **Export Engine:** Headless Chromium PDF renderer, jsPDF, XLSX.

Key Source Code Implementation Formulations:

1. Holt-Winters Smoothing Engine (`src/services/forecastEngine.ts`):

```
export function holtWintersMultiplicative(
  data: number[], seasonLength: number = 30, alpha = 0.2, beta = 0.1, gamma = 0.3, hori
): { forecast: number[]; lowerBound: number[]; upperBound: number[] } {
  let level = data.slice(0, seasonLength).reduce((a, b) => a + b, 0) / seasonLength;
  let trend = (data[seasonLength] - data[0]) / seasonLength;
  const seasonals = data.slice(0, seasonLength).map(v => v / (level || 1));

  for (let i = seasonLength; i < data.length; i++) {
    const lastLevel = level;
    level = alpha * (data[i] / (seasonals[i % seasonLength] || 1)) + (1 - alpha) * (lev
    trend = beta * (level - lastLevel) + (1 - beta) * trend;
    seasonals[i % seasonLength] = gamma * (data[i] / level) + (1 - gamma) * seasonals[i
  }
  const forecast: number[] = [];
  for (let h = 1; h <= horizon; h++) {
    forecast.push(Math.max(0, (level + h * trend) * seasonals[(data.length + h - 1) % s
  }
  return { forecast, ...computeConfidenceBands(forecast, data) };
}
```

2. Two-Phase Simplex LP Solver (`src/services/simplexOptimizer.ts`):

```
export function solveTwoPhaseSimplex(
  income: number, expenses: Record<ExpenseCategory, number>, targetSavings: number
): SimplexSolution {
  const targetCeiling = Math.max(0, income - targetSavings);
  const floors = calculateEssentialSurvivalFloors(income);

  // Phase 1: Check Feasibility
  const tableau = buildInitialTableau(expenses, floors, targetCeiling);
  if (!runSimplexPhase1(tableau)) {
    return { status: 'INFEASIBLE', message: 'Target savings breaches essential living f
  }
  // Phase 2: Compute Optimal Rebalanced Vector
  const optimalVector = runSimplexPhase2(tableau);
```

```
  return { status: 'OPTIMAL', allocations: optimalVector, iterations: tableau.iteration
}
```

REST API Routes:

Endpoint	Method	Request Payload	Response Action
`/api/v1/health`	GET	None	Health status, uptime, air-gap confirmation
`/api/v1/auth/login`	POST	`{ username, role }`	Role-based JWT session token (no password autofill)
`/api/v1/forecast/predict`	POST	`{ history: number[], horizon }`	30/90-day trajectory with 95% confidence intervals
`/api/v1/optimize/simplex`	POST	`{ income, expenses, savingsTarget }`	Optimal budget vector, category deltas, audit narrative
`/api/v1/compliance/mask-pii`	POST	`{ records: Transaction[], mask: true }`	Redacted legal records (`***-XX-****`)

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Figure 3: 30-Day Forward Cash-Flow Trajectory Chart with 95% Confidence Bounds

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Figure 4: Two-Phase Simplex LP Budget Rebalancing Interface & Variance Audit Chips

Results

The experimental evaluation was performed on 90-day longitudinal financial series and randomized budget envelopes across 8 spending categories.

1. Forecasting Model Comparison:

Forecasting Method	Mean Absolute Error (MAE)	Root Mean Square Error (RMSE)	Mean Absolute Percentage Error (MAPE)	95% Confidence Interval Coverage	Execution Latency
Holt-Winters Multiplicative	**₹842.10**	**₹1,248.50**	**4.12%**	**96.67%**	**0.84 ms**

Forecasting Method	Mean Absolute Error (MAE)	Root Mean Square Error (RMSE)	Mean Absolute Percentage Error (MAPE)	95% Confidence Interval Coverage	Execution Latency
14-Day Simple Moving Average	₹1,985.40	₹2,840.10	18.92%	73.33%	0.12 ms
30-Day Auto-Regressive AR(1)	₹1,420.30	₹1,950.80	11.45%	86.67%	3.20 ms

2. Two-Phase Simplex Optimization Performance:

- **Feasibility Detection Success Rate:** 100.0% (correctly rejected all budget envelopes violating essential living floors).
- **Mean Number of Pivot Iterations:** 3.8 iterations to reach global optimum.
- **Mean Execution Latency:** 3.42 milliseconds.

3. Anomaly Detection Performance (Z-Score ≥ 1.96 / > 35 Surge):

Metric	Observed Evaluation Value
Overall Classification Accuracy	98.66%
Anomaly Precision	70.0%
Anomaly Recall	87.5%
Anomaly F1-Score	77.8%

4. Case Study: Household Budget Rebalancing Under Expenditure Shock:

Budget Category	Initial Observed Baseline (₹)	Overspending Shock (₹)	Simplex Rebalanced Target (₹)	Rupee Adjustment
Monthly Net Salary	₹85,000	₹85,000	₹85,000	Baseline Income
Housing (Rent / EMI)	₹25,000 (29.4%)	₹25,000 (29.4%)	₹25,000 (29.4%)	Locked at Floor ($x_1 \geq ₹25,000$)
Food & Groceries	₹14,000 (16.5%)	₹19,500 (22.9%)	₹14,000 (16.5%)	Trimmed by $-₹5,500$ to healthy floor
Utilities & Bills	₹5,500 (6.5%)	₹6,500 (7.6%)	₹5,500 (6.5%)	Realigned to contractual floor
Healthcare	₹4,000 (4.7%)	₹4,000 (4.7%)	₹4,000 (4.7%)	Locked at Floor ($x_4 \geq ₹4,000$)

Budget Category	Initial Observed Baseline (₹)	Overspending Shock (₹)	Simplex Rebalanced Target (₹)	Rupee Adjustment
Shopping & Retail	₹8,000 (9.4%)	₹18,000 (21.2%)	₹6,000 (7.1%)	Elastic reduction of $-₹12,000$
Entertainment	₹6,000 (7.1%)	₹12,000 (14.1%)	₹3,000 (3.5%)	Elastic reduction of $-₹9,000$
Transport	₹5,000 (5.9%)	₹6,500 (7.6%)	₹4,500 (5.3%)	Realigned to transit baseline
Total Expenses	**₹67,500 (79.4%)**	**₹91,500 (107.6%) [Deficit]**	**₹62,000 (72.9%)**	**Total savings cut:** $-₹29,500$
Net Surplus / Savings	**+₹17,500 (20.6%)**	**-₹6,500 (Insolvent)**	**+₹23,000 (27.1%)**	**Solvency Guaranteed**

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Figure 5: Real-Time Anomaly Inspector and Outlier Variance Classification

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Figure 6: Executive Admin Reports Center with Toggleable PII Masking Shield

Conclusion:

In this mini-project, **FinPredict** was successfully designed, implemented, and evaluated as an institutional-grade decentralized Financial Planning and Analysis (FP&A) decision-support platform. By combining **Holt-Winters Multiplicative Triple Exponential Smoothing** for forward cash-flow forecasting with **Two-Phase Primal Simplex Linear Programming** for constrained budget rebalancing, the system provides deterministic, explainable mathematical recommendations that prevent personal and departmental insolvency.

The experimental results confirmed that Holt-Winters achieves high forecast precision ($\text{MAPE} = 4.12$, $\text{RMSE} = ₹1,248.50$), outperforming simple moving averages by 14.8%. The Two-Phase Simplex optimizer guarantees global optimality in $3.42\ \text{ms}$, ensuring essential living expenses are never compromised. The Zero-Trust air-gapped architecture, PII tokenization shield, Indian Rupee (`₹`) formatting, and enterprise MongoDB Atlas backend establish a robust, privacy-compliant solution aligned with NBA Program Outcomes (PO1, PO2, PO3, PO5, PO6) and UN Sustainable Development Goals 8 and 9.